

**PROCESS DESIGN AND OPERATING PHILOSOPHIES
ON
PRESSURE RELIEF & DISPOSAL SYSTEM**

OISD - STANDARD - 106

First Edition, November 1988
Amended edition, August, 1999

**Oil Industry Safety Directorate
Government of India
Ministry of Petroleum and Natural Gas**

ISD-STANDARD-106
First Edition
November 1988
Amended edition,
August, 1999

**FOR RESTRICTED
CIRCULATION**

**PROCESS DESIGN AND OPERATING PHILOSOPHIES
ON
PRESSURE RELIEF & DISPOSAL SYSTEM**

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Note ¹ in superscript indicates the changes / modifications / additions as approved in 17th Safety Council Meeting held in July, 1999.

FOREWORD

The oil industry in India is nearly 100 years old. As such a variety of practices have been in vogue because of collaboration/association with different foreign companies and governments. Standardisation in design philosophies, operating and maintenance practices at national level was hardly in existence. This, coupled with feed back from some serious accidents that occurred in the recent past in India and abroad, emphasised the need for the industry to review the existing state of art in designing, operating and maintaining oil and gas installations.

With this in view, the Ministry of Petroleum and Natural Gas in 1986 constituted a Safety Council in 1986, assisted by the Oil Industry Safety Directorate (OISD), staffed from within the industry in formulating and implementing a series of self-regulatory measures aimed at removing obsolescence, standardising and upgrading the existing standards to ensure safer operations. Accordingly OISD constituted a number of functional committees comprising of experts nominated from the industry to draw up standards and guidelines on various subjects.

The present document, on 'Pressure Relief & Disposal System' was prepared, by the Functional Committee on 'Process Design and Operating Philosophies'. This document is based on the accumulated knowledge and experience of industry members and the various national and international codes and practices. It is hoped that the provision of this document, if implemented objectively may go a long way to improve the safety and reduce accidents in the oil and gas industry. Suggestions are invited from the users for further improvement in the standard after it is put into practice. Suggestions for amendments to this document should be addressed to :

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PROCESS DESIGN AND OPERATING PHILOSOPHIES ON PRESSURE RELIEF & DISPOSAL SYSTEM

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PROCESS DESIGN AND OPERATING PHILOSOPHIES ON PRESSURE RELIEF & DISPOSAL SYSTEM

1.0 INTRODUCTION

The Pressure Relief and Disposal System is a key safety area in the hydrocarbon processing industries, API-520 (Part-1, 4th edition, 1976 & Part-II, 2nd edition, 1973) on Design and Installation of Pressure Relieving System in Refineries and API-521 (3rd edition, 1982), Guide for Pressure - Relieving and Depressurising Systems are well recognised documents and widely used in the petroleum industry all over the world. These guidelines are generally applicable to petroleum installations in India. However, it was felt necessary to modify certain provisions of these guidelines to conform to our climatic conditions, local practices and statutory requirements, and supplement with some provisions which are not addressed specifically in these guidelines.

2.0 SCOPE

The standard covers relieving devices and their discharge systems of vessels and equipment in the petroleum industry, which are designed for a maximum allowable working pressure of more than 1 Kg/Cm² g. This standard does not cover the atmospheric and low pressure tanks and pressure vessels used for transportation of petroleum products. The intent of the present standard is not to reproduce the above recommended practices, but highlight the areas of concern specific to our local environment and requirements.

3.0 DEFINITIONS

For the purpose of this standard the following definitions shall apply:

i) RELIEF VALVE

Is an automatic pressure-relieving device actuated by the static pressure on upstream of the valve.

ii) MAXIMUM ALLOWABLE WORKING PRESSURE:

It is the maximum gauge pressure permissible at the top of a vessel in its operating position for a designated temperature.

iii) OPERATING PRESSURE:

It is the gauge pressure to which the vessel is usually subjected in service.

iv) SET PRESSURE

It is the inlet pressure measured in gauge at which the pressure relief valve is adjusted to open/pop under service conditions.

v) OVER PRESSURE:

It is the gauge pressure on the discharge side of the safety valves.

4.0 NEED FOR RELIEVING SYSTEM

The relieving of pressure from a process system arises from a number of reasons as below:

- i) This may be required so that a system is not allowed to pressurise beyond its maximum allowable working pressure, in order to avoid possible failure of the weakest part of the system.
- ii) For precautionary relieving of pressure from the system called as depressurisation. This is applicable for high pressure and or high inventory systems which need to be depressurised during an emergency.
- iii) In the case of a fire, the maximum allowable yield stress of the metal reduces significantly due to increased temperature. Relieving pressure under these situations allows the actual stresses to be reduced below the lowered maximum allowable stresses thereby preventing failure.
- iv) To take care of thermal expansions when a pipeline or equipment containing a liquid is blocked in and subsequently heated.

5.0 PRESSURE RELIEVING / SAFETY DEVICES

There are basically two type of safety devices used for relieving pressure in a system. These are — Safety/Relief valves and Rupture Discs. Safety/Relief Valves may be conventional type, balanced bellow type & pilot operated type.

5.1 SAFETY/RELIEF VALVES

5.1.1 Conventional

Conventional safety/relief valves are susceptible to both superimposed and built-up back pressure and are not recommended when the total back pressure exceeds 10% of the set pressure. For these reactions, these should be used only in system relieving to atmosphere like steam, air or other non-toxic and non-flammable materials.

5.1.2 Balanced Bellows

Balanced Bellows valves are not susceptible to back pressure and should be used for back pressure upto 50% of set pressure.

5.1.3 Pilot-operated

In pilot-operated safety valves, the main safety valve opens through a pilot valve. Both the pilot and the main valve contain flexible membranes, which can withstand only ordinary service temperatures. Because of this and the risk of fouling, their use is limited to very clean services and are generally not recommended in hydrocarbon services.

5.2 RUPTURE DISCS

Rupture Discs are thin metal diaphragms held between flanges and are designed to burst at the set pressure. Once burst, these are not reusable and have to be replaced. Their set pressure cannot be tested without destroying them. After the test, the rupture disc has to be replaced but there is no guarantee that the second rupture disc will burst at the same pressure. This is a major disadvantage of rupture disc, especially when the bursting pressure is low. For these reasons, rupture discs alone shall not be used. However, they should be used between the vessel and a relief/safety valve for fluid of highly corrosive or fouling nature. Prolonged expose of safety valve directly to such conditions may cause damage to valve components.

5.3 SET PRESSURE OF RELIEF VALVES

- (a) Relief valves should usually be set at 10% higher over the normal operating pressure to allow a reasonable margin so that the valves do not open frequently with minor process upsets. The difference between the set pressure and the normal operating pressure should not be less than 2 Kg/CM². This is

to account for operating contingencies and the fact that spring setting of safety valve at lower pressure is not of high precision. This aspect should be considered for selecting the design pressure (maximum allowable working pressure) of the equipment. The design pressure or maximum allowable working pressure is the highest pressure at which the pressure relief is set to open.

- (b) The safety valve set pressure in trunk pipelines should be set within 10% above the maximum allowable operating pressure.
- (c) When rupture disc is used, the bursting pressure of the rupture disc should be kept 5% lower than the safety valve set pressure. In order to have a reasonable margin between the bursting pressure and the normal operating pressure, the relief valve set pressure should be 15% higher than the normal operating pressure. A pressure gauge/bleeder between rupture disc and relief valve helps to indicate the health of the rupture disc.

6.0 INSTALLATION OF SAFETY DEVICES

6.1 GENERAL

Relieving devices should be installed directly on the equipment they are protecting or immediately adjacent on the connected piping without any valve in the piping. These devices are best installed on the top of vessels or at high points so as to minimise and simplify the inlet piping. Following guidelines shall apply:

- (a) Inlet piping shall be adequately sized so as to limit pressure drop between vessel and safety valve to 3% of the set pressure on the inlet side.
- (b) The discharge side including the header shall be sized so as to contain total back pressure within permissible limits depending upon the type of safety valve.
- (c) Inlet and outlet of a safety valve shall not be less than the nominal sizes of inlet/outlet flanges respectively of the safety valve.
- (d) Inlet and outlet (if pressure relieving device is discharging to a closed system) piping shall be free draining away from the safety valve.
- (e) The discharge line shall join the header from top and directionally to avoid high pressure drop.
- (f) In vessels where there are chances of liquid carryover alongwith vapour in the form of froth, mist, etc., the inlet line to safety valve and the outlet line

from safety valve to the unit knock-out/Blowdown drum shall be sized based on two-phase flow.

6.2 MULTIPLE VALVES

IBR code requires that that total relieving capacity of a system must be broken up into at least two safety valves. Multiple valves may also be required for hydro-carbon processing services, if the total relieving capacity required cannot be provided in a single valve. Multiple relief valves should not be provided with isolation valves. Installation of multiple valves allows staggered set points for each valve as recommended by API & ASME. When the required relieving capacity is provided in more than one pressure-relieving device, only one device need be set at the maximum allowable working pressure, and the additional devices may be set at higher pressures but in no case at a pressure higher than 105% of the maximum allowable working pressure. If, however, the pressure relieving devices are used for protection against fire or other sources of external heat, the additional devices can be set at a pressure not exceeding 110% of the maximum allowable working pressure.

Multiple relief valves with staggered set pressures are as such recommended to increase life of the valves and minimise leakage through the valves. A small leak due to operating pressure reaching above the set pressure will cause greater leakage in large valves. This, apart from leakage, also leads to wear and reduced life of valves.

6.3 SPARE SAFETY VALVES

Spare safety valves are often installed to facilitate testing and maintenance of one safety valve while the other is on line. As per Static and Mobile Pressure Vessels (unfired) Rules, 1981, every pressure vessel used for storage of compressed gases including liquefied petroleum gases should be provided with two or more pressure relieving devices.

Further, these are also used for continuity of operation in case of safety valve does not reseal after popping. Under such situations, isolation valves on the inlet and outlet of each relief valve are installed with some provision for keeping the isolation valves in open position (carseal or others).

Though this practice is quite common, in use, there are some inherent risks. Inclusion of isolation valves increases the number of flanges and total piping in the system, leading to increased possibility of leakage's, inadvertent inclusion of blinds, etc. The worst case is

inadvertent closing of isolation valves on both the safety valves. Chances of slip blinds remaining in position during construction and testing of the system are more when there are more number of flanges. Checking of such a system before start up becomes more difficult and requires extra care.

Therefore, spare safety valves should not be installed unless absolutely necessary or are required by a statutory authority. A detailed examination of the service conditions of the system should be done and if the conditions are very critical, only then spare safety valves should be installed. If spare safety valves are absolutely necessary, three way isolation valves should be installed at the inlet and outlet with proper manifolding. Alternatively, single isolation valve upstream and downstream of each safety valve shall be provided. These isolation valves should preferably be installed with their stem pointing downwards to avoid the possibility of a valve remaining stuck closed in case the stem becomes free. Sometime, isolation valve upstream of a single safety valve is provided to facilitate inspection and maintenance. This practice is not safe, as the valve may get inadvertently closed or when the valve is closed for isolating the safety valve, the system will remain unprotected.

7.0 CALCULATION OF RELIEVING LOADS

7.1 INDIVIDUAL LOADS

API-520 Part-1, Section-4 gives guidelines for estimating the relieving capacity for each safety valve under different contingencies that may occur in a plant. While determining individual relieving loads, following key points should be kept in mind:

- (a) Every piece of equipment that can generate a vapour or liquid load under any contingency must be recognised after doing a detailed analysis. No load should be left unconsidered for being small.
- (b) A pressure relief valve handling a liquid at vapour equilibrium or a mixed-phase fluid will produce flashing with vapour generation as the fluid moves through the valve. This may reduce the effective mass flow capacity of the valve and must be taken into account. Section 3.17.1 of API-521, 2nd edition, should be referred to for estimating the loads of safety valves under such conditions.
- (c) While calculating the load for a safety valve under fire condition, following key points should be considered:
 - (i) No credit should be taken for the insulation provided on the vessel.
 - (ii) No credit should be taken of safety devices such as shutdown switches, solenoid valves,

etc., Such devices should be assumed to fail in the case of a fire. If, on the other hand, a positive action of safety device (e.g. emergency steam into heater coils) will add to the relief load, it would be assumed to function.

7.2 GROUPING OF RELIEVING LOADS

7.2.1 Plant-wise

The individual loads estimated as above should be grouped together for various contingencies in order to design the relieving system components downstream of the safety valves. A table listing such loads should be prepared for each plant or facility in the complex. From this table the total governing load for the largest single contingency for each plant should be estimated.

7.2.2 Complex-wise

In a big complex, where a number of individual plants and facilities are connected to a common relief system, the relieving load for the entire complex has to be estimated considering the relieving loads for individual plant or facilities as mentioned earlier. Grouping of the individual plant loads should be done very judiciously based on the utility system design of the complex.

For example, if a common cooling water system serves a number of plants, the relieving loads from all such plants should be added together in case of cooling water failure.

If, however, there are more than one cooling water system in the complex each fed by independent reliable power supply, the failure of one cooling water system may call for grouping the relieving loads from only those plants which are served by this cooling water system.

It is important that a complete analysis of various contingencies that may occur in the complex be done and their overall effect recognised very carefully while estimating the relieving loads for the entire complex.

8.0 DISPOSAL SYSTEM

The purpose of a disposal system is to conduct the relieved gas or liquid to a safe location where it does not pose any hazard to human life, property or to environment. In some situations, the relieved vapors may safely be discharged to atmosphere directly. But in many situations where the fluid relieved is toxic or form a flammable mixture in air, the same should be disposed of through an appropriately designed closed disposal system. Such a system may consist of vessels, pipes, pumps, flare, vent scrubber and incinerator etc. and the associated

instrument and control auxiliaries. The choice of an appropriate disposal system will depend on the nature of relieved fluid and other local conditions.

8.1 ATMOSPHERIC DISCHARGE

General

Atmospheric discharge of relieved vapours is the simplex of all the disposal methods. However, there are many hazards while handling flammable and toxic vapours. The decision to discharge such vapours to atmosphere requires careful attention to ensure that disposal can be accomplished without creating a potential hazard or causing other problems, such as the formation of flammable mixture at grade level or on elevated structures, exposure of personnel to toxic vapours or corrosive chemicals, ignition of relieved stream at the point of emission, excessive noise levels and air pollution.

It is recommended that no hydrocarbon and other toxic releases be discharged to a atmosphere directly,. However, in certain situation like marketing installations, LPG bottling plants and other remotely located installations where hydrocarbons are stores and handled and no flare or other closed disposal systems are feasible, the relieved vapours may be discharged to atmosphere. Following key points should be kept in min under these situations:

- (a) The relief valves should discharge to atmosphere vertically upwards through their own individual stacks, so sized that minimum exit velocity of 150 meter/sec would be obtained. The maximum velocity should be well below the sonic velocity and should not exceed 0.5 mach. If feasible, snuffing steam should be connected to the vents. under these conditions, the air entertainment rate is very high and the released gases will then be diluted to below their lower flammable limit.
- (b) A single common vent stack should not be used for several relief valves because this results in a discharge velocity much less than the designed discharge velocity when only one safety valve is operating.
- (c) The vent stack should discharge at a minimum elevation of 3 meters above grade or the tallest structure, within a radius of 15 meters, whichever is higher.
- (d) Individual vents should have a drain hole of 1/2" at the low point in the vent line. The drain connection should be piped to a safe location.
- (e) If the relieved vapours will produce excessive noise at the nearest operating structure, the vent line/stack

should be provided with acoustic insulation. Silencers should not be used as they are likely to block the outlet due to fouling, etc.

8.2 CLOSED DISPOSAL SYSTEM

All hydrocarbons, toxic vapour and liquid releases should be discharged through a closed disposal system like flare, vent scrubber, incinerator or a blowdown drum as the case may be. The primary function of a flare or incinerator is to convert flammable, toxic or corrosive vapours to less objectionable compounds by combustion.

Toxic vapours like SO₂, phenol, chlorine, etc. which cannot be converted to less objectionable compounds by combustion should be disposed of through a vent scrubber using caustic soda, water or other suitable agents as the case may be. Refer Section 8.5 for details of scrubbers.

8.3 DESIGN OF CLOSED DISPOSAL SYSTEM

Once of the various combinations of loads have been defined for all pertinent contingencies and the corresponding allowable back pressure have been determined for all relief valves, selection and design of various components of the disposal system can proceed as below:

8.3.1 Gathering Network

Individual relief valve discharges and other vents should be combined and piped to a flare or a vent scrubber as the case may be. All laterals and headers shall be free draining away from the safety valves towards a knock-out drum with a minimum slope of 1 in 500. All laterals should join the header from above. Flare header should be continuously purged from any convenient location to avoid air ingress in to the system. Fuel gas, inert gas and nitrogen are commonly used as purge material. Steam should not be used as a purge material because it condenses in the system and pulls vacuum. Purge rates should be between 0.05 ft/sec. to 0.1 ft/sec. as measured at the flare stack for flare systems having gas seals at the flare stack tip.

The network should have adequate expansion loops to account for the temperature range of the released material. Where viscous materials are handled which solidify on cooling at the ambient temperatures, the lines should be heat traced.

8.3.2 Unit Isolation Valve

Large complexes have many units feeding to a common flare system. Since units must be isolated from rest of the complex for maintenance, isolation valves with blinds at the battery limits of units should be considered. These isolation valves should be installed with their stems pointing downwards so as to minimise the chances of accidental closure. Butterfly valves should not be used as isolation valves.

8.3.3 Unit Knock-out Drum

Wherever the discharge from a unit is expected to contain appreciable quantities of liquids, especially corrosive, fouling and congealing in nature, a Knock-out (K.O) drum of suitable size must be installed at the battery limit of each such unit with flare line sloping towards the K.O. drum. Unit K.O. drum may also be required if the layout of the units is such that it is not feasible to have a continuous sloping of the flare header(s) towards the main flare K.O drum. The liquid collected in these drums should be drained/pumped to a suitable disposal system like a closed blowdown drum or slop system and not to open drains, while the vapours led to the flare header.

These drums should be sized to separate particles of 300 - 600 micron size and designed to hold the liquid discharge expected for 5 - 10 minutes from a single contingency. Gravity draining of liquid from these drums to the unit closed blowdown drum etc. should be preferred. Alternatively, two pumps each sized to empty out the drum hold up in 15--20 minutes should be provided. These pumps should start/stop automatically at high/low level in the drum respectively.

These drums also should be provided with High and low level alarms and level indicator in the control room of these unit. If a congealing type of liquid is likely to be handled, these drums should be heat traced or provided with steam coils.

8.3.4 Cryogenic Discharges

Cold liquid and vapour discharges pose additional problems of metallurgy. Such releases should be handled separately before they join the main flare header. Liquid discharges at sub-zero temperature should be piped to a separate drum provided with suitable vapourising system. Care should be taken that the heating medium, usually steam, does not get frozen by the extreme cold. If it is not feasible to vapourise/heat the cold discharged material, the entire flare piping design should be of suitable material compatible with the service conditions. Under such situations it may be more economical to segregate cryogenic discharges from the main flare header.

8.3.5 Main Flare Header

The main flare header collects the material relieved through safety valves for safe discharge to the flare stack for combustion. If unit K.O. drums are provided the flare headers downstream need be sized only for vapour flow. It may be economical to have more than one main flare header if there is substantial difference between the allowable back pressures of different safety valves. The flare header should be so sized that the back pressure at the outlet of any safety valve does not exceed the max. possible value.

The flare header should not have any pocket and be free draining towards the nearest K.O. drum. A slope of 1 in 500 is normally recommended. No check valves should be permitted in the flare header system.

If the liquids to be handled include oil with a relatively high pour point, provision should be made to avoid solidification in the system. Likewise, the introduction of high viscosity oils may require protection against low ambient temperatures, particularly on instrument leads. use of heat tracing is recommended under such situations H₂S is corrosive and if handled together with the main flare header, may lead to corrosion of the same. It is recommended to have a separate flare header preferably of stainless steel for handling H₂S.

Sizing of flare headers is usually done on pressure drop considerations. However, a check should be made to ensure that the maximum velocity in the header is well below the sonic velocity. A value of 0.2 Mach (maximum) is recommended. In many situations, the emergency discharge is at high temperatures, and flare header runs many hundred meters. This results in the loss of flare gas temperature due to heat loss to metal and surroundings. In order to estimate the total pressure drop in the flare header, the total header length should be divided into a number of small sections say 100 meters or so and pressure drop in each such section should be estimated taking into account the change in vapour density in each section.

8.3.6 Main Flare Knock-out Drum

In addition to the unit K.O drums, a main flare K.O. drum should be installed close to the flare stack. This takes care of any liquids condensed due to atmospheric cooling of the headers.

Horizontal and vertical drums are both acceptable. Due to high vapour flow rates, split flow horizontal drums are usually economical. The drums should be sized to

separate out liquid driplets of 300-600 microns size. Heavy entrainment of liquid may lead to fire balls from the flare stack falling on the grade which can lead to serious consequences. The K.O. drums should be sized to provide liquid hold up of 20-30 minutes, with one meter hold up in vertical vessels and 1/4 dia in horizontal vessels.

Two pumps, one running and the standby should be

provided to pump out the liquid collected in the K.O drum to a safe location. Pump capacity should be such that the liquid hold up can be emptied out in 15-20 minutes. Provision of emergency drive (steam turbine or alternate source of power) should be considered so that these pumps can be operated during the failure of normal power supply. The pumps should be designed to start automatically on high liquid level and stop on low level. It shall also be possible to start/stop these pumps from remote control room. The extent of instrumentation should be same as mentioned in 8.3.3

Selection of internals for the K.O. drum should be given a careful consideration. Internals that may clog or otherwise foul up should not be used. Vortex breakers should be used in the liquid outlet lines. Adequate arrangements should be made to handle congealing liquids. Heating coils in the K.O. drum, tracing of liquid lines with steam or electric tracers should be considered. Under these circumstances, K.O. drums should be sized for a min. design pressure of 4-5 Kg/cm² a.

8.3.7 Seal drums

Seal Drums together with provision for purging and the installation of flare seals provide adequate protection against flash back from the flare tip. These drums are usually vertical and should be mounted as close to the flare stack as possible. Seal drums integral with flare stack are also commonly used.

Refer fig.1 for details of a typical seal drum. Some use flame arrests in the flare header close to the stack to guard against any flash back. Such devices are likely to get blocked or fouled up resulting in higher back pressures in the flare header. Sometimes they may completely block the header. Also their inspection is very difficult. For these reasons, flame arrestors should not be used in the flare system.

The seal drum shall have a cross sectional area equal to 4 times the inlet pipe dia and be designed for 4.5 Kg./CM² as minimum. The inlet pipe should rise vertically for at least 3 meters from the water level to avoid ingress

of air in to the system due to vacuum created when hot vapours cool off. Enough water must be stored in the vessel so that seal does not break under such conditions. This might necessitate an increase of the drum dia. Maximum allowable back pressure in the header will decide the maximum submergence of inlet pipe under the seal. A minimum seal of 100 mm is recommended. Maximum seal height should not exceed 300 mm to avoid puffing of flame.

Water shall be continuously added to the seal drum and the overflow shall be automatic through a liquid seal leg. As a minimum, the leg height should be equal to 1.75 times the maximum expected operating pressure (not design pressure). All lines connecting K.O. drums, seal drums and the flare stack shall be free of pockets. The seal leg should be provided with a 1½" siphon breaker. Provision should be made to skim off any oil that may accumulate in water seal drum.

In cold places where there is possibility of freezing of water in the water seal drum, a steam coil should be provided in the drum to keep water warmed up. A level gauge, and high a low level alarms also should be provided on the seal Drum.

8.3.8 Flare Stack

Flare stack are usually elevated structures designed to burn out flammable vapours safely so as to cause minimal damage to environment, population and property. Such flare stacks are usually associated with a certain amount of smoke, noise and glare which are considered to be public nuisances.

Of late, ground, box or enclosed flares have become popular to minimise public nuisance problems. These flares are, however, complex pieces of heavily instrumented equipment. These should be used to burn off completely only the normal flaring loads which are usually small. These shall not be used as means of disposal of emergency loads. The box flare load should never exceed its rated capacity during emergency. These must always be backed up by elevated stacks which automatically take over the emergency loads. The switching of loads between the elevated and box flares should be accomplished by means of appropriate water seals.

Smokeless flaring may also be achieved by use of steam and proprietary flare tips.

Flare stack diameter is usually based on maximum allowable velocity which should be considered as 0.2 Mach for normal loads and 0.5 Mach for short time emergency loads. Flare stacks should also be provided with gas seals to prevent flash back and cutdown the purge gas rates.

The stack height is based on the maximum allowable radiation level at the nearest location which may be ground level or other elevated structures. Table-3 of API-521, 3rd edition, list acceptable radiation levels for personnel exposed to radiation. The radiation levels given in this table are exclusive of solar radiation. However, for our country, where solar radiation is comparatively much higher (about 300 BTU/hr. ft²) the radiation level as given in API-521, 3rd edition, should be considered inclusive of solar radiation. In some situations (for example cold flaring), ground level concentrations of flare gas may govern the height of stack. All the applicable pollution standards should be followed while finalising the height of flare stack.

The flare stack should be located at a safe distance from plant and storage area and also from public roads and property. Ref. Standard OISD-118 on "Plant Layouts" for such distances. When two flares are operating, the separation distance between the two must be checked from allowable radiation consideration as discussed earlier.

It is necessary to have sterile area around the flare stack for a sufficient radius, free of grass and other growth to avoid any fire hazard by falling of burning liquid from the stack.

8.3.9 Pilot Ignitors

To ensure ignition of flare gases, continuous pilots with a means of remote ignition are recommended for all flares. The most commonly used type of ignitor is the flame-front propagation type, which use a spark from a remote location to ignite a flammable mixture. Ignitor control panel should be located away from the base of elevated flares. For the box flares, the ignitor panel should be located at least 15M away from the box flare. The panel should be provided with a canopy to protect men and equipment from liquid spill or thermal radiation.

8.3.10 Standby Flare

Various units, storage and handling facilities of a complex may be connected to a single flare. It must be recognised that flare stack and the associated auxiliaries do require some inspection and maintenance, for which

these may not be available. If all the process units, storage, handling and other facilities which are connected to the flare system are not shutdown together and some facilities are operative, a standby flare with appropriate isolation devices should be provided. Alternatively, the entire load from a complex should be suitably distributed among two or more flares so that each one of them can be inspected/repaired during partial shutdown of the complex.

8.4 DISPOSAL OF HEAVY LIQUIDS AND CONDENSABLE VAPOURS

8.4.1 General

If a unit has a sizable amount of relieving load consisting of a heavy and fouling type of liquids and condensable vapours, it may be desirable to have a separate disposal system dedicated to such safety valves which discharge liquids and/or condensable vapours. In addition to the provision of section 8.1 on "Atmospheric Discharge" or the provision of Section 8.3 on design of "Closed Disposal System", as the case may be, the following guidelines should apply for disposal under these situations:

8.4.2 Cold and Heavy Hydrocarbon Discharge

Heavy Hydrocarbons which are not expected to vapourise at atmospheric pressures and operating temperatures must be discharged through a closed system to a blowdown drum. See figure-2 for details of such a system. If the hydrocarbons are highly viscous or would solidify at ambient temperatures, the piping, valves, etc. in the system should be heat traced. The blowdown drum should be sized to hold the largest liquid relief for 5-10 minutes. The liquid from the drum can be pumped to a slop system. The design criteria for pumps and the instrument details should be similar to that mentioned under 8.3.3 on "Unit Knock-out Drums". The blowdown drum should be connected to flare. A steam coil and temperature indicator should be provided in the drum, if the liquid discharged is heavy and congealing type. This blowdown drum can be combined with the closed drain system of the units. Under these situations, the drum should be located underground and vented to atmosphere with a steam purge. See fig.-3.

8.4.3 Hot and Heavy Hydrocarbons Discharges

Heavy hydrocarbons, which, because of high temperatures, might be expected to evolve a large amount of vapour, should be discharged to a quench drum. Such a system with relevant details is depicted in figure-4. The

quench drum is a vertical vessel fitted with baffles and is connected by means of a conical transition to the flare header. The hot hydrocarbon material is fed in to the drum below the baffles. Suitable quenching medium like water or gas oil etc. is sprayed at the top of baffles under temperature control. The cooled hydrocarbon liquid along with the quench material is drawn from the bottom of the drum and disposed of to sewer.

The uncondensed vapours along with any steam formed, pass up the quench drum to the flare header. The vapour line from the drum to the flare should be sized properly to take care of any steam formed.

8.4.4 Thermal Relief Discharges

When piping, vessels and exchangers are blocked in with cold liquids in them and are subsequently heated by heat tracing or other means of heat input, hydraulic expansion takes place which can cause serious failures. A thermal relief valve usually of $\frac{3}{4} \times 1$ " nominal size should be used to take care of this phenomenon. TSVs may be relieved to a closed blowdown vessel in process units and in closed/open blowdown system in offsite. ^{Note1}

8.4.5 Discharge from Relief Valves of Pumps

Discharge of liquids from safety/relief valves on the discharges of pumps should be returned to the suction line or suction vessel from which the pump takes suction.

8.5 DISPOSAL OF TOXIC/CORROSIVE MATERIALS

8.5.1 General

If the relieved material which is to be discharged is of toxic or corrosive nature and does not burn effectively in a flare, such materials should be disposed of after scrubbing/neutralising thoroughly.

Such disposal systems typically use a neutralising agent either as a large pool or spray in a contact tower. The type of relieved material involved and the choice of scrubbing/neutralising agent and disposal system should be considered from case to case. Table-1 gives a list of some typical relieved materials along with the type of disposal system.

8.5.2 Design Considerations

From consideration of metallurgy to handle corrosive materials and to reduce the load on the scrubbing system, the

discharge of toxic and corrosive chemicals should be

pipled separately from other hydrocarbon discharges. The capacity of the system should take care of the largest release of the toxic materials.

(a) Neutralisation in Pool

In a scrubbing system consisting of a pool of appropriate liquid, the discharged vapours should pass in to the liquid pool through a well designed sparger supported at the bottom of liquid pool. The liquid level in the pool containing neutralising agent should always be maintained. Whenever the liquid strength gets exhausted after a release, the liquid should be replaced.

(b) Neutralisation in a Spray Tower

The disposal system may consist of a spray tower instead of a pool. Such a tower is provided with baffles. The scrubbing/neutralising liquid is circulated from the bottom of the tower to the top by means of a pump. Any vapour released in to the spray tower will react with circulating liquid before getting released to flare/atmosphere. See figure-5 for a typical spray tower, it should be active so that the system and so designed as to have minimal dependence on utilities and instrumentation. For example, if power failure is the cause of emergency, the disposal system shall not use pumps driven by electric power. Pools of neutralising agent or overhead storage of an adequate amount of neutralising agent can be considered in such cases. Attention shall be paid to the back pressure allowable in the disposal system.

9.0 VAPOUR DEPRESSURISING

9.1 RUNAWAY REACTIONS

In some processes like hydrocracking, reforming and oxidation process there is a likelihood of a runaway reaction leading to sudden rise in system pressure. Further, some processes may be susceptible to frequent pressure surges. Safety relief valves may start leaking if used to take care of such situations which may occur frequently. In such situations emergency depressurising systems are recommended to be used in addition to relief valves. The depressurising system drops the system pressure rapidly through a pressure control valve. Such discharges should be lead to the closed disposal system like flared, etc.

Off gases containing oxygen from vessels shall not be routed to flare. Example bitumen blowing unit. ^{Note 1}

9.2 EXPOSURE TO FIRE

When a vessel under pressure is exposed to fire, the metal temperature may reach a level at which stress rupture of the vessel could occur, even though the pressure may not go beyond the relief valve set pressure. Emergency depressurising systems are recommended for rapidly removing vapours from vessels exposed to a fire. A typical vapour-depressurising system should reduce the pressure in the vessel to 50% of the design pressure in 15 minutes. Following three effects should be considered while estimating the amount of vapours to be handles by the emergency depressurising system:

- Vapours generated from liquid by heat input from fire.
- A change in density of internal vapour due to reduction on pressure.
- Liquid flash due to pressure reduction, when the system contains liquids at its saturation temperature.

9.3 DISPOSAL OF MATERIAL FROM DEPRESSURISING

The vapours from hydrocarbon emergency depressurising system should be routed to the closed disposal system as described in Section 8.0 Depressurising valves are generally 100% open or 100% closed. The maximum load from an open depressurising valve will thus correspond to the flow capacity of the control valve at the maximum pressure of the protected equipment.

10.0 SAFETY/RELIEF VALVE DATA SHEET

It is important that all the relevant process data for the pressure relieving devices be furnished in the form of a standard data sheet. The Process Engineer must examine the various conditions that may occur in the process before firming up this data. A sample sheet of Safety/Relief valve data sheet is given as Annexure-I

11.0 RELIEF SYSTEM RECORD

A record of all the relief valves and the other components of the relief system e.g. headers, pump, vessels, etc. should be kept in the plant. This record should be reviewed before undertaking any modifications in the plant facilities which are likely to increase the relieving load from the plant. Such a review of the relief

system is important when increase in plant capacity, major changes in operating conditions or addition of some equipment for better energy recovery or other considerations is being planned. From safety considerations, it is imperative that sizing calculations for pressure relieving and disposal devices should be made a part of permanent plant record. A sample sheet showing the record of a safety valve is given here as Annexure-II. Records for other components of the relief system should also be made on similar lines.

12.0 REFERENCES

1. API 520. Recommended Practice for the Design and Installation of Pressure-Relieving Systems in Refineries, Part-1, Design Section 4, 4th edition 1976 and Part II, 2nd edition 1973.
2. API 521. Guide for Pressure-Relieving and Depressuring System, Section 4, 3rd edition 1982.
3. Static and Mobile Pressure Vessels (unfired) Rules 1981, Chapter-ii
4. OISD-118, Layout of Petroleum Installations Section 7, 1st edition 1988.

Table - 1**TOXIC/CORROSIVE CHEMICALS**

Relieved Material	Scrubbing/Neutralising Agent	Type of Treatment
1. Sulphur Oxides	Soln. of lime or soda ash	Scrubbing
2. Ammonia	— Water	Dilution with large quantity of Water.
3. Chlorine	Soln. of soda ash or line	Absorption in large pool of alkali.
4. Phenols	Alkaline water (pH=8.5)	Scrubbing
5. Hydrogen Sulphide	Dilute Caustic Solution	Scrubbing
6. Furfural	Water	Scrubbing
7. Tetraethyl lead (TEL)	Kerosene	Absorption in pool of water
8. Glycols	Water	Absorption in pool of water
9. Nitrogen Oxides	Soln. of caustic soda	Scrubbing
10. Sulpholine	Water	Scrubbing
11. Amines	Water	Scrubbing
12. MEK	Water	Scrubbing

[illegible]

ANNEXURE II

TAG NO. _____

SAFETY RELIEF VALVE RECORD

Vessels or Equipment Protected _____

CODE: ASME POWER BOILER/ASME UNFIRED PRESSURE VESSELS/API RP 520

Operating PR _____ Relieving PR _____ Accumulation _____

Design PR: _____

Constant Back Pressure _____ Variable Back Pressure _____ (MAX)

Spring Set Pressure _____ Valve Type: Conventional/Bellows _____

Fluid _____ (Vapour/Liquid) sp.gr./Mol.Wt _____

Latent Heat _____ Tempo-Operating _____ Relieving _____

Special Requirements _____

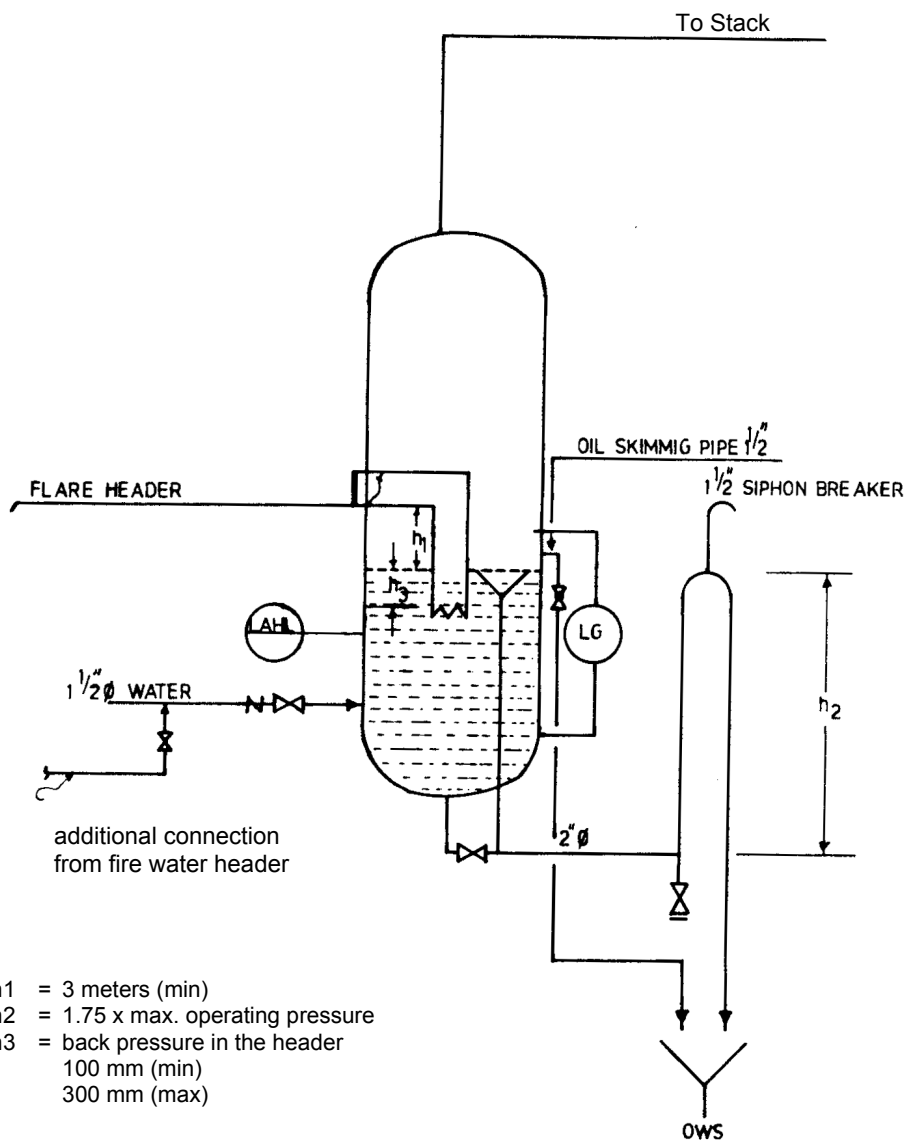
BASIS FOR SIZING

CONDITION	REQD.CAPACITY	CONDITION	REQD.CAPACITY
Fire	_____	Power Failure	_____
Block Outlet	_____	Cooling Water Loss	_____
Tube Failure	_____	Reflux Loss	_____
Control Failure	_____	Liquid Expansion	_____

REQUIRED AREA

As Per API 520:.....

Area of Valve Selected.....



legend

lg : level gauge
lahl : level alarms high level
ows : oily water sewer
O : panel instrument
O : local instrument

fig. : 1
water seal drum

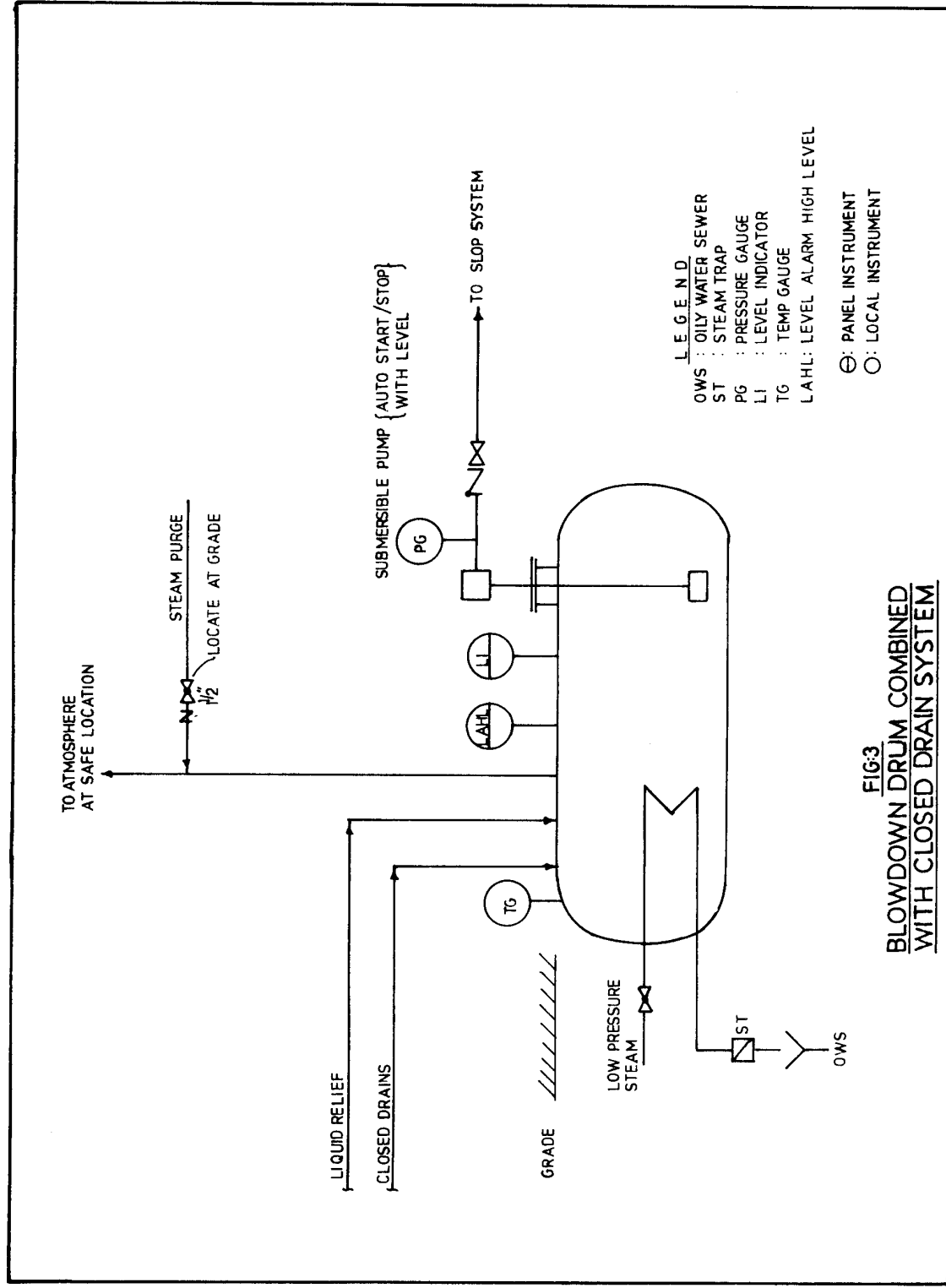
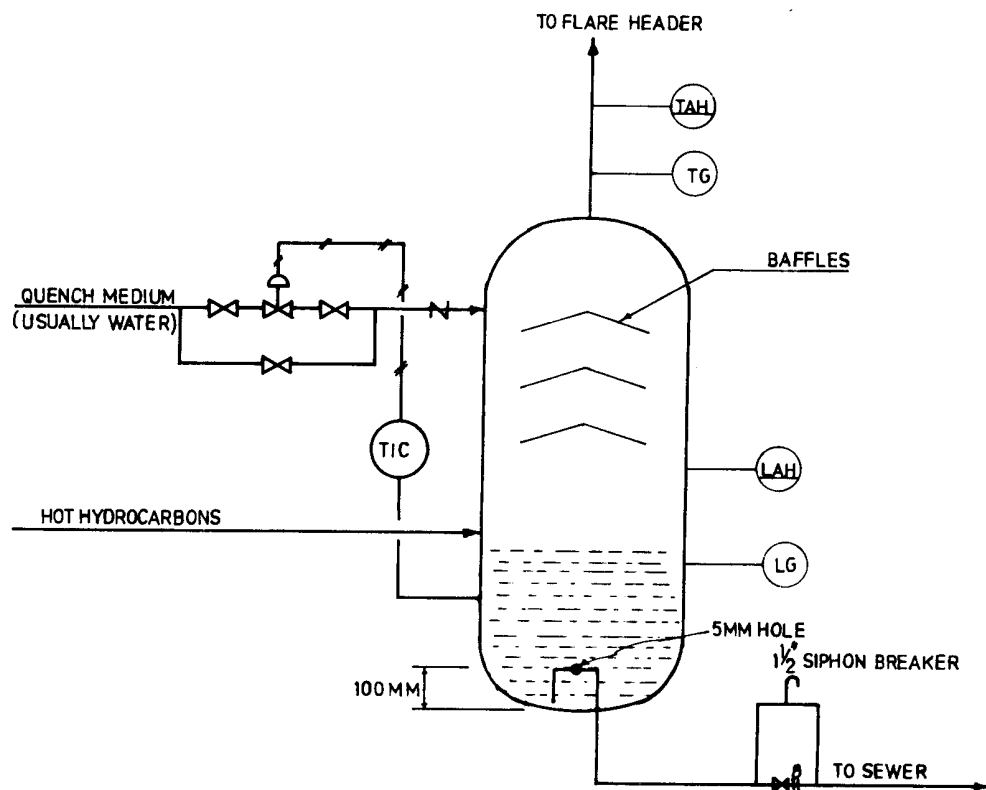


FIG:3
**BLOWDOWN DRUM COMBINED
WITH CLOSED DRAIN SYSTEM**



LEGEND

TIC : TEMPERATURE INDICATOR & CONTROLLER
 LG : LEVEL GAUGE
 TG : TEMP GAUGE
 LAH: LEVEL ALARM HIGH
 TAH: TEMP ALARM HIGH
 ⊖ : PANEL INSTRUMENT
 ○ : LOCAL INSTRUMENT

FIG:4
QUENCH DRUM

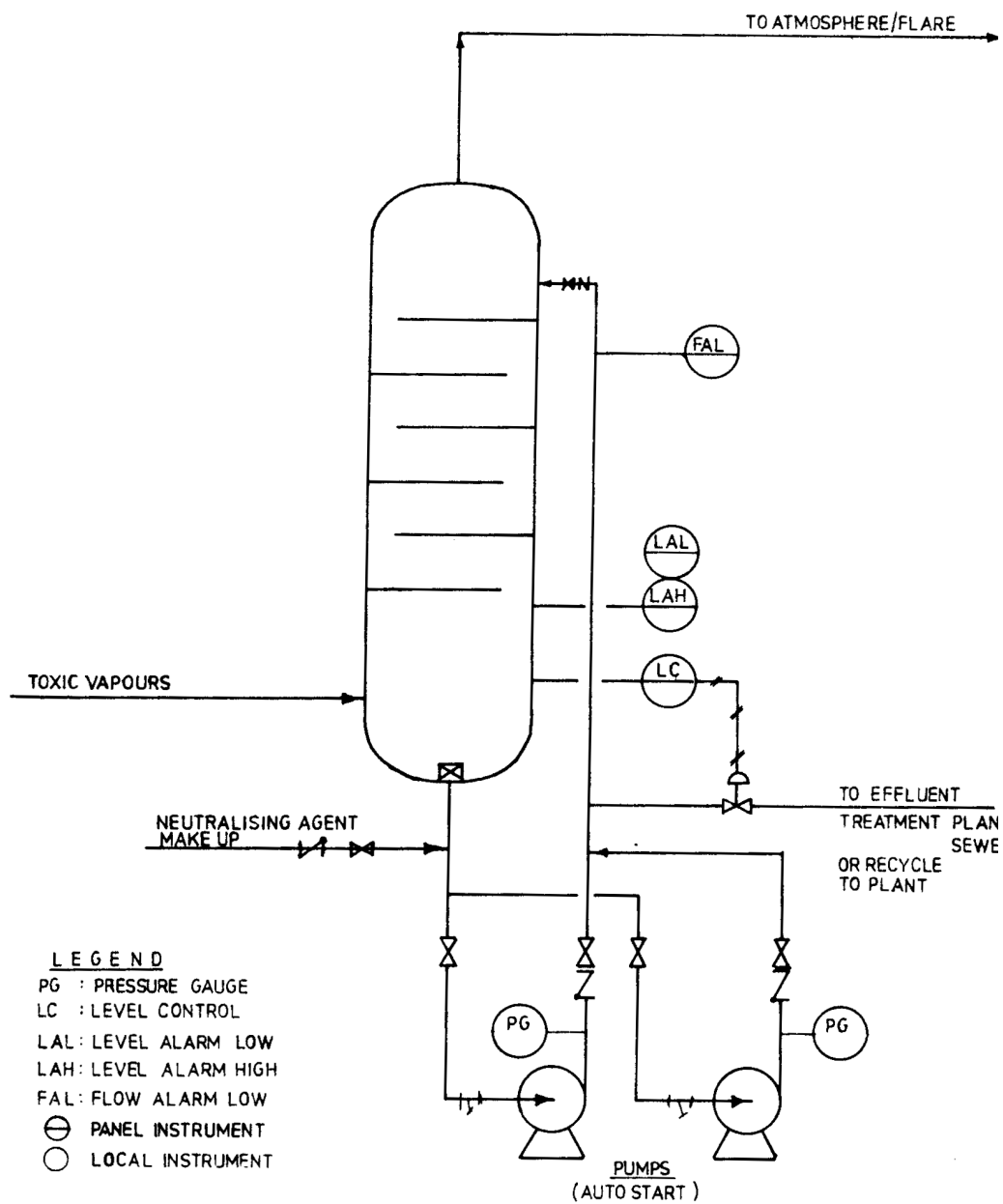


FIG:5
SCRUBBING UNIT

